

REVIEW ARTICLE

Advances in Multiple Sclerosis

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SUMMARY

Multiple sclerosis is a chronic autoimmune disorder that affects the central nervous system, causing episodes of neurologic dysfunction and often gradual disease progression. The immune system primarily targets myelin, the protective covering of nerve fibers, leading to inflammation and damage, and secondary neurodegeneration is a major cause of long-term disability. Common symptoms include vision problems, sensory disturbances, muscle weakness, balance difficulties, and bladder dysfunction. Important advances in treatment have improved outcomes in patients with relapsing forms of multiple sclerosis, particularly through highly effective immune-modifying therapies such as CD20-targeting monoclonal antibodies. However, treatment options for progressive forms remain limited, which highlights the need for therapies that can prevent progression and promote myelin repair. Comprehensive symptom management and lifestyle support are also essential to maintaining quality of life and reducing disability.

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CME



MULTIPLE SCLEROSIS, A COMMON AUTOIMMUNE DISEASE OF THE CENTRAL nervous system, affects the brain, optic nerves, and spinal cord. Its hallmark effects are inflammation, demyelination, and neurodegeneration. In traditional descriptions of the disease, the myelin sheaths surrounding nerve-cell axons are the main target; peripheral nerve myelin, although structurally similar, is unaffected.

The disease typically manifests with discrete focal attacks that affect vision, sensation, strength, and balance, and over time the frequency of attacks decreases and progressive worsening with prominent neuronal injury supervenes (Fig. 1). In a minority of patients, symptoms are progressive from onset. Diagnosis rests on clinical criteria and magnetic resonance imaging (MRI) findings, supplemented by clonal expansions of immunoglobulin in cerebrospinal fluid and slowing in central nerve conduction velocities as measured by evoked potentials.

Etiologic features of multiple sclerosis include a complex genetic cause, a predisposing Epstein-Barr virus infection, and an immune response centered on B and T lymphocytes and microglia. Highly effective therapies, such as CD20-depleting monoclonal antibodies, provide near-complete protection against relapses and partial protection against progressive symptoms. The achievement of greater control of progression and underlying neurodegeneration is the most important unmet need in the field.

ORIGINS

Multiple sclerosis was first formally recognized in the mid-1800s. Its incidence appears to have increased during the past century, even when improved diagnosis is taken into account.¹ In the United States, nearly 1 million people have multiple sclerosis, with women affected nearly three times as often as men.²

Although both genetic and environmental factors are implicated in risk, this observed increase in incidence points to some environmental change.

GENETICS

Complex polygenic factors underlie the development of multiple sclerosis. Concordance in monozygotic twins is approximately 25%, and siblings of affected persons have a 3% lifetime risk of the disease. More than 230 risk-associated single-nucleotide polymorphisms (SNPs) have been identified, nearly all of which are of small effect (odds ratios of 1.1 to 1.3); most are intronic (i.e., not within gene coding regions).³ Functional and pathway studies indicate that numerous variants promote the activation and responsiveness of the immune system to inflammatory stimuli.⁴ The strongest genomewide locus is located within the major histocompatibility complex on chromosome 6, with more than 30 independent signals in this region, and with the class 2 major histocompatibility complex gene *DRB1*15:01* and its extended haplotype carrying the greatest risk. Multiple sclerosis alleles tend to cluster with genome maps for other autoimmune diseases and not with neurodegenerative or other brain diseases (Fig. 2). Almost all multiple sclerosis risk variants have little influence on the disease course, lack specificity and sensitivity, and thus have no role in diagnosis or management. One locus that is associated with disease severity is in the region encoding dysferlin and a zinc finger protein (*ZNF638*); homozygous carriers have accelerated motor disability.⁶

EPSTEIN–BARR VIRUS AND OTHER ENVIRONMENTAL TRIGGERS

Accumulating evidence suggests that multiple sclerosis develops, at least in part, as a rare complication of Epstein–Barr virus infection.⁷ Multiple sclerosis is rare in persons never exposed to Epstein–Barr virus, and initial exposure to the virus in adolescence or adulthood, rather than earlier in life, is associated with infectious mononucleosis, higher titers of antibodies to viral proteins, and heightened risk.⁸ Of note, acute infection with Epstein–Barr virus is usually followed by lifelong infection, almost exclusively carried in memory B cells, a finding that could help explain how the disease develops and the success of B-cell–directed therapies. Multiple sclerosis risk is also weakly associated with vitamin D defi-

ciency, obesity, smoking, and certain gut bacteria, including *Akkermansia muciniphila*.⁹

NEUROPATHOLOGICAL FEATURES

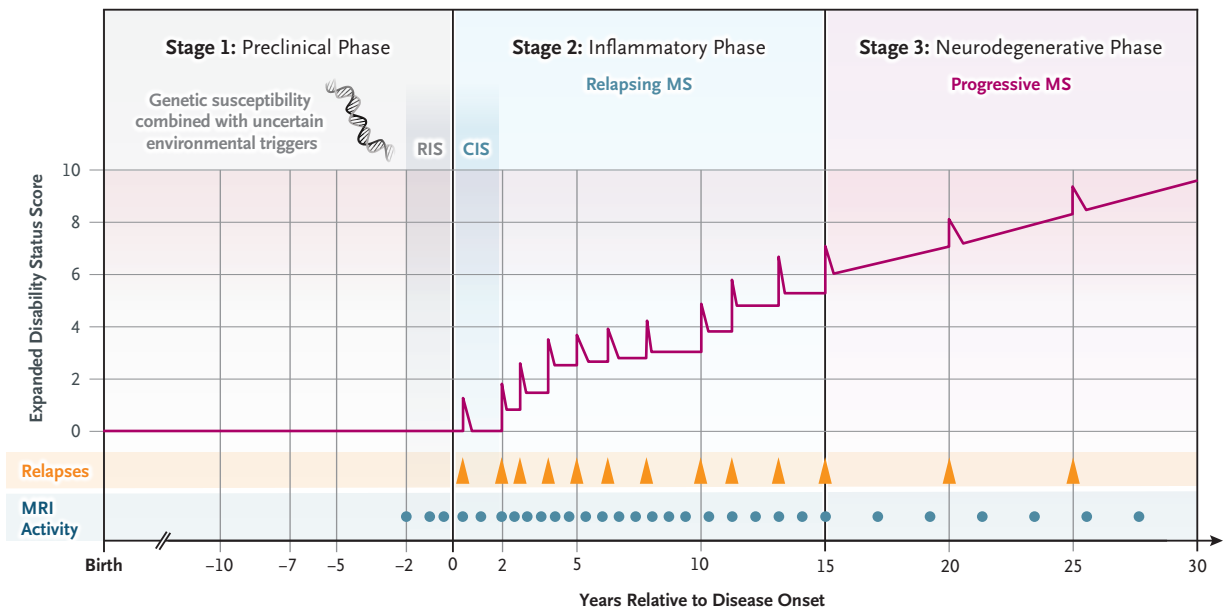
Several patterns of inflammation and tissue damage occur over the life span of the illness (Fig. 3A).¹⁸ Clinical relapses are caused by acute focal lesions, whereas all pathological features contribute to progressive symptoms.

Acute focal lesions are characterized by perivenular disruption of the blood–brain barrier, infiltration by mononuclear cells (B and T lymphocytes, natural killer cells, and monocytes), demyelination, axonal damage, and loss of oligodendrocytes, the myelin-producing cells of the central nervous system.¹⁹ These lesions are often most evident in the myelin-rich regions of the cerebral hemispheres (cerebral white matter), optic nerves, and spinal cord, but also occur in the cerebral cortex and other gray-matter areas that contain only sparse myelin.

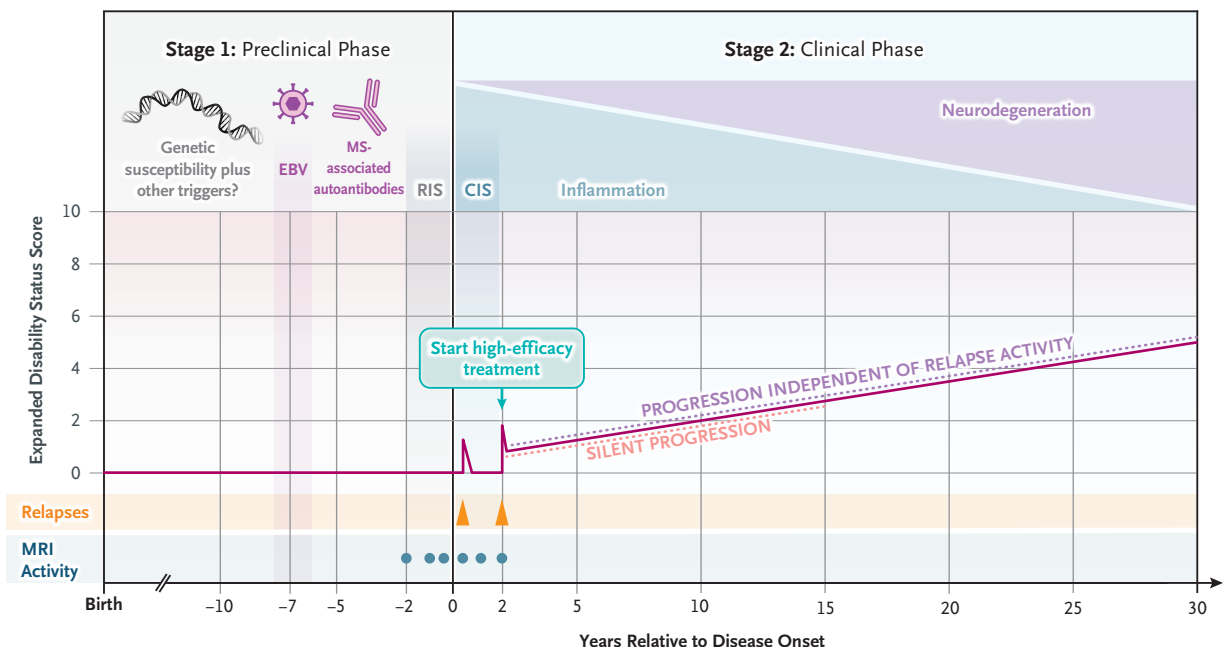
Acute inflammation typically subsides over several weeks, leaving sharply demarcated astrocytic scars with various degrees of remyelination (plaques). Many lesions manifest with no ongoing inflammation (chronic inactive lesions). In other lesions, however, inflammation becomes chronic (chronic active lesions), with prominent microglia and only sparse lymphocytes; some chronic active lesions also gradually enlarge over time.²⁰ Paramagnetic rim lesions (defined by a rim of increased magnetic susceptibility on MRI sequences) are particularly destructive lesions with activated iron-containing microglia at the advancing border and associated with accelerated axonal injury; they are specific to multiple sclerosis and associate with greater clinical severity.²¹ A distinguishing feature of chronic active lesions, in contrast to acute focal lesions, is that the blood–brain barrier is relatively intact owing to the reestablishment of endothelial tight junctions associated with resolution of perivascular inflammation.

Prominent neuronal loss and demyelination within the cerebral cortex is another key pathological aspect of the disease, even in early cases.²² Although some cortical lesions are caused by intracortical plaques, attention has focused on a distinctive injury associated with overlying ectopic lymphoid follicles composed of aggregates of B, T, and plasmablast cells located in the meninges adjacent to the pia mater and often overlying

A The Traditional Model



B New Concepts in the Current Treatment Era



deep sulci.²³ The close proximity of meningeal lymphoid aggregates with damage to underlying cortical tissue suggests that secreted factors — cytokines, free radicals, or antibodies — could be responsible for cortical involvement. In long-

standing and severe cases, extensive changes include diffuse microgliosis and astrogliosis, variable infiltration of CD8 T cells, widespread myelin pallor, axon loss, synaptic injury, and global brain and spinal cord atrophy.²⁴

Figure 1 (facing page). Evolving Understanding of Multiple Sclerosis.

Panel A shows traditional concepts of the natural history of relapsing multiple sclerosis (MS) before the era of highly effective treatment. The purple line indicates disease progression. Disease onset begins after a pre-clinical phase, with environmental risk factors acting on a genetically susceptible person. Magnetic resonance imaging (MRI) evidence of preclinical MS is occasionally identified in asymptomatic persons (radiologically isolated syndrome [RIS]). The initial attack is designated as a clinically isolated syndrome (CIS). In the traditional model, during the relapsing phase, disability accumulation was thought to result entirely from incomplete recovery from relapses, until clinically significant relapse-independent disability (secondary progressive multiple sclerosis) supervened, on average 15 to 16 years after clinical onset. Less commonly, progressive MS developed without earlier relapses (primary progressive multiple sclerosis; not shown). Panel B summarizes more recent concepts regarding the natural history of MS in the current treatment era. A key role of Epstein–Barr virus (EBV) infection has been established during the preclinical phase, and specific serum autoantibodies cross-reactive to EBV and other proteins are present in some patients. With early treatment, attacks are prevented in most patients, but insidious progression independent of relapse activity, known as silent progression (or progression independent of disease activity), is now evident during the relapsing phase, which reveal that both inflammation and neurodegeneration are present throughout the disease course. In addition, the time to reach permanent disability milestones and secondary progressive multiple sclerosis is substantially lengthened. Scores on the Expanded Disability Status Scale range from 0 to 10, with higher scores indicating worse disability.

IMMUNOPATHOGENESIS

ADAPTIVE IMMUNITY

The success of B-cell–targeted therapies for all forms of multiple sclerosis has focused attention on the central role of the humoral immune system in this disease (Fig. 3B). B-cell invasion in the central nervous system is associated with more severe tissue damage and a more aggressive clinical course.²³ Expanded populations of clonally restricted memory B cells in the central nervous system produce oligoclonal immunoglobulins (oligoclonal bands) in the cerebrospinal fluid, long used as a diagnostic marker in multiple sclerosis.²⁵ Although most antibodies target non-specific intracellular proteins,²⁶ others are also found deposited on degenerating myelin sheaths in multiple sclerosis lesions.²⁷ Reactivity has been reported against numerous central nervous system targets including myelin oligodendrocyte

glycoprotein, myelin basic protein, glial-cell adhesion molecule, anoctamin-2, and $\alpha\beta$ crystallin.²⁸

B cells are also efficient antigen-presenting cells, and B-cell mediated activation of pathogenic T cells probably plays a role in multiple sclerosis.^{11,12} Interactions between B and T cells could also account for the rapid onset of action of B-cell–depleting therapies, which reduce brain inflammation long before downstream changes in antibody levels occur. Expansions of CD4 T cells that are reactive against myelin proteins have been repeatedly identified.²⁹ Cytotoxic CD8 cells are the predominant T-cell subset at the leading edge of lesions, and clonally expanded CD8 cells have also been shown to react against relevant myelin antigens.³⁰

In studies examining the role of Epstein–Barr virus in multiple sclerosis,³¹ investigators have found that Epstein–Barr infection of B cells can promote migration to the central nervous system,³² survival, autoantigen presentation to T cells,³³ and autoantibody production.³¹ Cerebrospinal fluid immunoglobulins and some oligoclonal bands react against Epstein–Barr proteins, including nuclear antigens (EBNA1 and EBNA3) and the activation-associated proteins LMP1 and BRRF2.³⁴ Several antibodies that are reactive to Epstein–Barr virus and that are enriched in multiple sclerosis have been found to cross-react with normal human proteins in the central nervous system. Some of these antibodies are directed to a motif of BRRF2 and are present in the serum of some patients years before symptoms begin and thus represent a prodromal marker of multiple sclerosis risk.³⁵ Antigen-experienced CD4 and CD8 T cells that react against Epstein–Barr virus have also been identified, some of which cross-react with brain antigens.¹²

INNATE IMMUNITY

Activated microglia are prominent in chronic active plaques and are associated with progressive injury to neurons, axons, and oligodendrocytes. In patients with long-standing disease, widespread diffuse microgliosis throughout the white matter can be observed. Injurious proinflammatory molecules released by microglia include tumor necrosis factor α , interleukin-1, and complement protein 1q. These, along with interleukin-6 and granulocyte–macrophage colony-stimulating factor (GM-CSF) released by B and T lymphocytes, promote an activated pathogenic astrocyte response with reduced buffering capacity

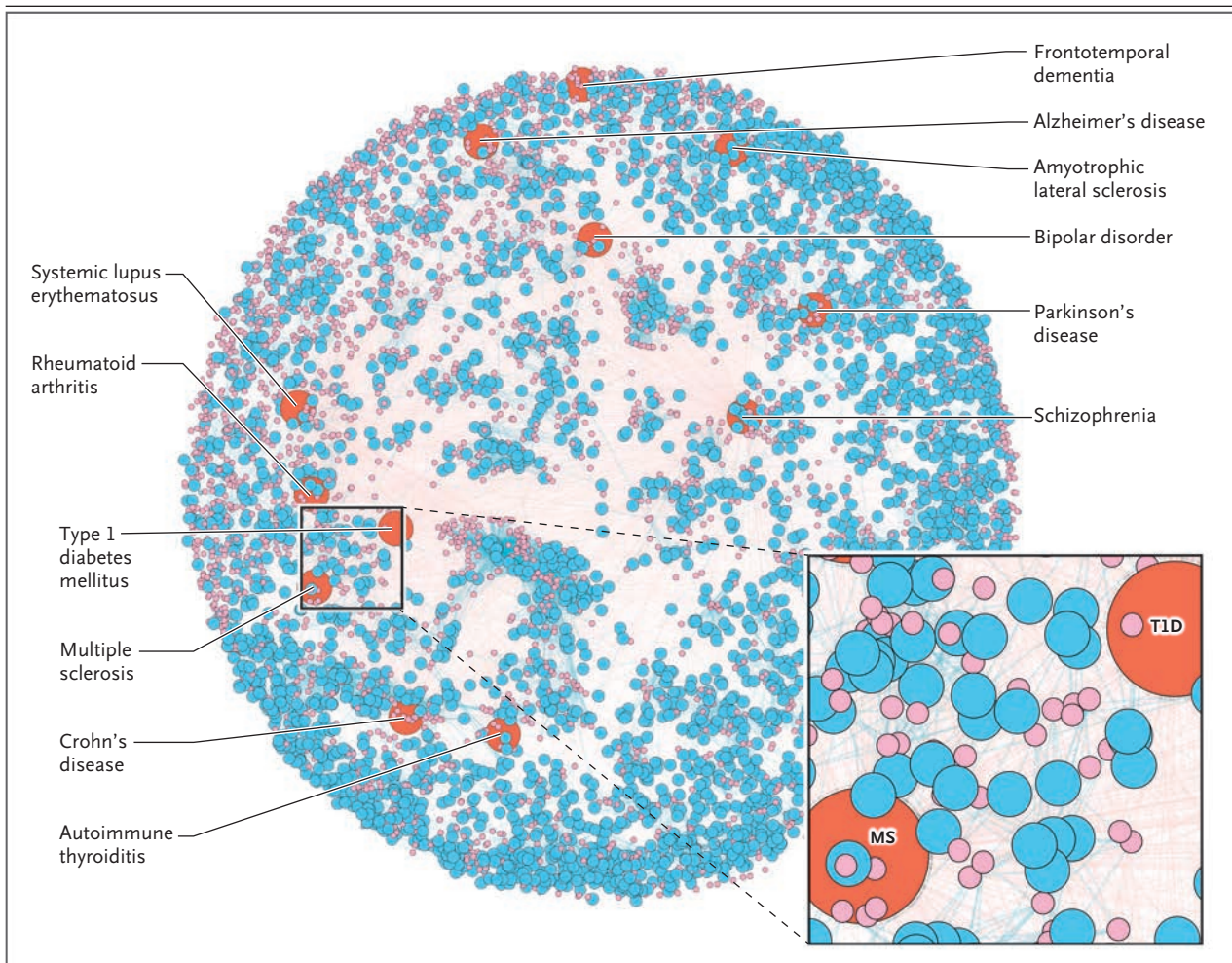


Figure 2. Multiple Sclerosis Genetics Landscape.

Genetic susceptibility for multiple sclerosis clusters with other autoimmune diseases, but not with neurodegenerative diseases. Genetic associations for 12 diseases (6 neurodegenerative and 6 autoimmune) were extracted from the SPOKE (Scalable Precision Medicine Open Knowledge Engine) knowledge graph (which obtains genetic information from the Genome-Wide Association Study Catalog).⁵ Each disease (large red circles) is connected to its associated single-nucleotide polymorphisms (SNPs; small pink circles) by pink lines (shown in expanded view, inset), overall representing the genetic architecture of all 12 diseases. Diseases and SNPs were clustered with the use of the Cytoscape software spring-embedded layout algorithm, which positions diseases next to one another on the basis of shared SNP associations. In a subsequent step, genes (blue circles) were arranged with the same layout algorithm on the basis of the SNPs they contain within their extended boundaries (connected by blue lines, inset). The strongest genomewide locus for multiple sclerosis lies within the major histocompatibility complex on chromosome 6. T1D denotes type 1 diabetes mellitus. Data visualization courtesy of Sergio Baranzini.

for neurotoxic glutamate and loss of normal homeostatic control of synaptogenesis.³⁶

CLINICAL FEATURES

DETECTION

Symptoms are typically first noted in early adulthood, although onset as early as 2 years of age or as late as the eighth decade of life can occur. Monocular vision loss (optic neuritis) or diplopia

(double vision), sensory changes, weakness, imbalance, vertigo, and urinary bladder disorders are typical early symptoms. Fatigue, heat sensitivity, mental clouding, and electric-shock sensations evoked by neck flexion (Lhermitte's sign) are also common. New symptoms — relapses or attacks — can appear over hours to days and then remit over weeks to several months. Early in the course of the disease, remissions are often robust, but as the disease becomes chronic, the

likelihood for recovery lessens. In approximately 10% of cases, the disease begins as progressive worsening without attacks — primary progressive multiple sclerosis — a subtype that has a nearly equal incidence in women and men.

Diagnosis is often suspected from the history and is supported when the neurologic examination reveals signs of central nervous system injury, sparing the peripheral nervous system. MRI typically reveals multiple sharply demarcated lesions in the periventricular, juxtacortical, and deep white matter of the cerebral hemispheres and in the spinal cord. The presence of a central vein in some lesions and paramagnetic rim lesions³⁷ can be diagnostically helpful (Fig. 4). Acute inflammatory lesions are identified by extravasation of gadolinium dye indicating disruption of the blood–brain barrier; lesions tend to be enhanced on imaging for approximately 1 month and then fade, and nearly all leave a permanent visible scar. In early cases, approximately 90% of acute focal lesions are asymptomatic, but accumulation of these clinically silent lesions over time contributes to progressive disability.³⁸

Lumbar puncture can be an important adjunctive test. Characteristic findings in the cerebrospinal fluid include increases in immunoglobulin levels and oligoclonal bands, the presence of kappa free light chains, modest inflammation with mononuclear cells, and normal levels of protein and glucose. Evoked potentials measure conduction velocity in visual, auditory, sensory, and motor pathways; delays, especially when asymmetric, can indicate demyelination. Elevated levels of neuron-specific neurofilament light chain and the astrocyte marker glial fibrillary acidic protein (GFAP) can be detected in plasma or serum and indicate nervous system injury,³⁹ but at the current time these biomarkers have only limited use as guides to diagnosis or treatment of individual patients.

RED FLAGS

Systemic symptoms should point to the possibility of a different autoimmune disorder or a vascular or neoplastic cause, such as sarcoidosis, systemic lupus erythematosus, small-vessel stroke, vasculitis, or lymphoma. A multiple sclerosis mimic is more likely in patients who present with solitary lesions or pure progressive syndromes. In all patients, vitamin B12 deficiency should be excluded. When severe optic nerve and spinal cord presentations predominate, autoimmune

disorders that were previously considered to be variants of multiple sclerosis but now are recognized as distinct disorders need to be considered; these include neuromyelitis optica, myelin oligodendrocyte glycoprotein antibody–associated disease, and GFAP disease. Many cases of acute disseminated encephalomyelitis can now be reclassified as myelin oligodendrocyte glycoprotein antibody–associated disease. Although the clinician should always be alert to the possibility of an alternate diagnosis, in the proper clinical context, a correct diagnosis of multiple sclerosis is usually clear-cut.

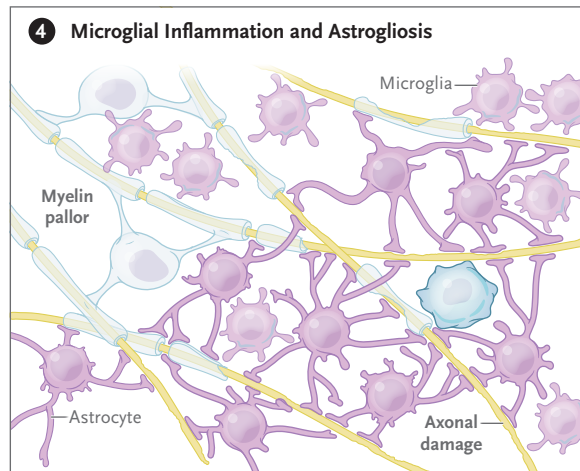
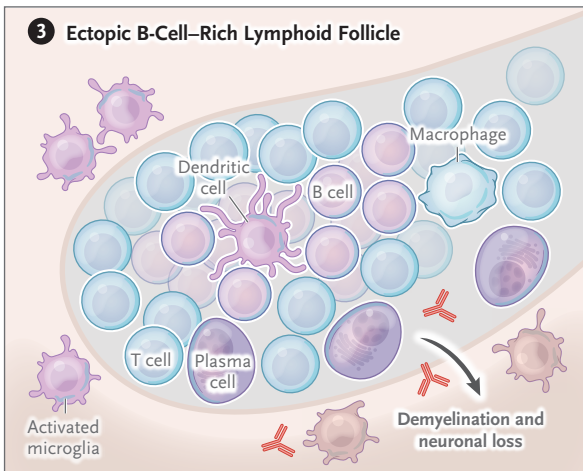
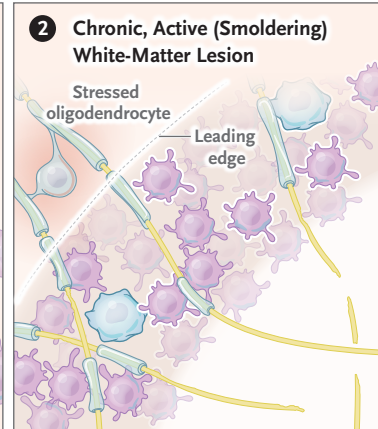
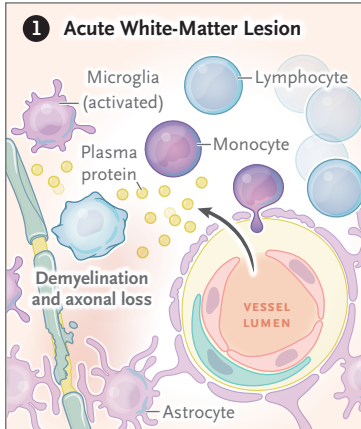
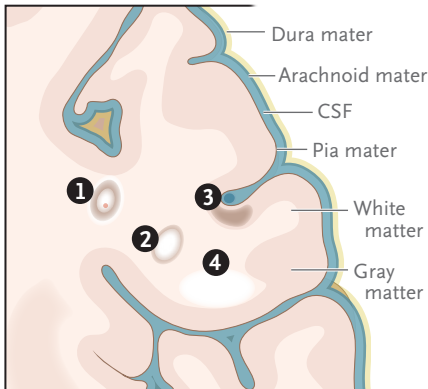
DIAGNOSIS

Diagnostic criteria⁴⁰ require the presence of two or more episodes, each lasting more than 24 hours and separated by at least one month, plus abnormalities on examination that indicate localization to more than one site. In patients who are having a first-ever clinical attack (clinically isolated syndrome), multiple sclerosis can be diagnosed if the MRI documents both an enhancing and a nonenhancing (e.g., older) lesion. In patients with only a single abnormality on examination, either MRI or evoked responses can be used to identify a noncontiguous clinically silent abnormality. In patients who present with progressive symptoms and gradual worsening over a period of more than 6 months, MRI findings or cerebrospinal changes can support the diagnosis. A diagnosis is also possible in asymptomatic patients for whom incidental MRI findings are suggestive of multiple sclerosis if those findings are supported by additional MRI-based criteria or cerebrospinal fluid changes.

COURSE

Traditional concepts stratify multiple sclerosis into three principal types, distinguished by whether predominant symptoms are attributable to attacks or to gradual progression. These designations — relapsing–remitting, secondary progressive (which follows an earlier relapsing–remitting course), and primary progressive — are still useful in the approach to treatment, and therapies continue to be approved by regulators on the basis of these classifications. However, these phenotypes represent a single disease entity, defined by identical genetics and epidemiologic and pathologic features.⁴¹ The development of effective therapies for relapsing multiple sclerosis also brought to light a silent

A Pathologic Features



B Immune Mechanisms

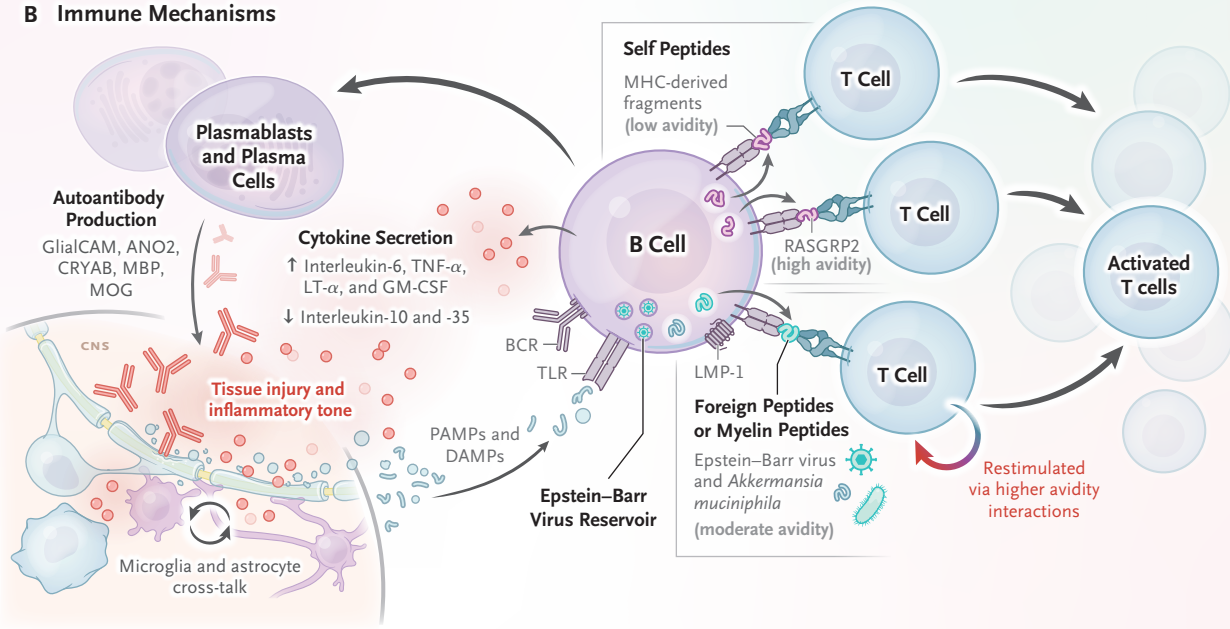


Figure 3 (facing page). Immunopathogenesis of Multiple Sclerosis.

Multiple pathologic features acting over the life span of the illness contribute to relapses, progression, and disability (Panel A). Acute focal inflammatory white-matter lesions are responsible for relapses but are more often asymptomatic; hallmarks are perivenous inflammation with lymphocytes and monocytes, disruption of the blood–brain barrier with leakage of plasma proteins, demyelination associated with activated microglia, astrocyte proliferation (gliosis), and axonal loss with secondary retrograde and anterograde degeneration. Acute lesions can evolve to become slowly enlarging (“smoldering”) chronic active lesions, some with a rim of activated microglia at the leading edge (paramagnetic rim lesions), stressed oligodendrocytes, and progressive axonal injury; other acute lesions become inactive, and in some, partial remyelination can occur. B-cell-rich lymphoid aggregates form in the deep sulci of the meninges, producing underlying cortical demyelination and neuronal loss. In long-standing cases, widespread microglial inflammation and astrogliosis throughout the nervous system is present, associated with myelin pallor and ongoing axonal damage; these changes can be superimposed on age-related neurodegeneration. B cells can contribute to tissue damage in multiple sclerosis through varied mechanisms (Panel B). Antigen presentation of self-peptide fragments derived from major histocompatibility complex (MHC) molecules and other proteins expressed in B cells trigger low avidity activation of a pool of responding T cells, which can be restimulated through higher avidity interactions with multiple sclerosis–associated viral and bacterial components as well as cross-reactive myelin antigens.^{10–12} Polarized B cells expressing higher than normal levels of disease-associated cytokines (interleukin 6, tumor necrosis factor- α [TNF- α], and lymphotoxin- α [LT- α]) and lower levels of regulatory molecules (interleukin-10 and interleukin-35) promote inflammation.¹³ Autoantibodies targeted to myelin and other components of the central nervous system, produced by B-cell–derived plasmablasts and plasma cells, may also contribute to tissue injury. B cells receive feedback from T cells, in particular T follicular helper cells, which are prominent in the nervous system of persons with multiple sclerosis.¹⁴ Toll-like receptors expressed on B cells can recognize damage-associated signals (pathogen-associated molecular pattern molecules [PAMPs] and damage-associated molecular pattern molecules [DAMPs]), further stimulating inflammation. There is extensive cross-talk between microglia and astrocytes,¹⁵ as well as between B and T lymphocytes.¹⁶ Many cell–cell interactions amplify damage, while others promote resilience and repair.¹⁷ B cells are also the long-term reservoir of Epstein–Barr virus persistence, and B cells infected with Epstein–Barr virus are thought to play a catalytic role in these processes. ANO2 denotes anoctamin-2, CRYAB $\alpha\beta$ crystallin, CSF cerebrospinal fluid, GlialCAM glial-cell adhesion molecule, MBP myelin basic protein, MOG myelin oligodendrocyte glycoprotein, and RASGRP2 RAS guanyl-releasing protein 2.

progression (also known as progression independent of relapse activity) that is present in most or all patients⁴² and was previously obscured by a fluctuating neurologic status caused by episodic relapses followed by gradual recovery of various degrees (Fig. 1). Insidious progression, rather than relapse-associated injury, represents the most important cause of permanent neurologic disability in patients with relapsing–remitting multiple sclerosis.⁴³ The term relapsing (as distinguished from relapsing–remitting) multiple sclerosis bundles together patients with relapsing–remitting and secondary progressive disease who continue to have attacks. In practical terms, secondary progressive multiple sclerosis can be applied to patients with relapsing–remitting disease who have substantial walking disability not entirely caused by relapses.⁴⁴

In the era before efficacious therapies for relapse prevention were available, approximately half the patients with relapsing–remitting multiple sclerosis could be expected to enter a secondary progressive course, with substantial motor disability, within 15 years after onset of disease, and only 20% had no functional limitations. After widespread adoption of efficacious therapies beginning in the 1990s, an improved long-term prognosis became evident, with a mean time of approximately 40 years until the development of secondary progressive multiple sclerosis.⁴⁵ Although this advance was probably due, at least in part, to benefits of treatment, a change in the underlying natural history of the disease may also have occurred. With widespread use of therapies with higher efficacy over the past decade, an even more favorable long-term outlook has emerged. As examples, in open-label extension trials, after 10 years of continuous treatment with ocrelizumab, more than three fourths of patients with relapsing multiple sclerosis had no sustained disease worsening,⁴⁶ and nearly 95% of patients who received ofatumumab therapy had no disease activity after 5 years of treatment.⁴⁷ In these two trials, improvement from baseline was more common than worsening.

TREATMENT

ACUTE RELAPSES

Multiple sclerosis relapses, including initial attacks, can be treated with high-dose glucocorticoids for 3 to 5 days, sometimes followed by a

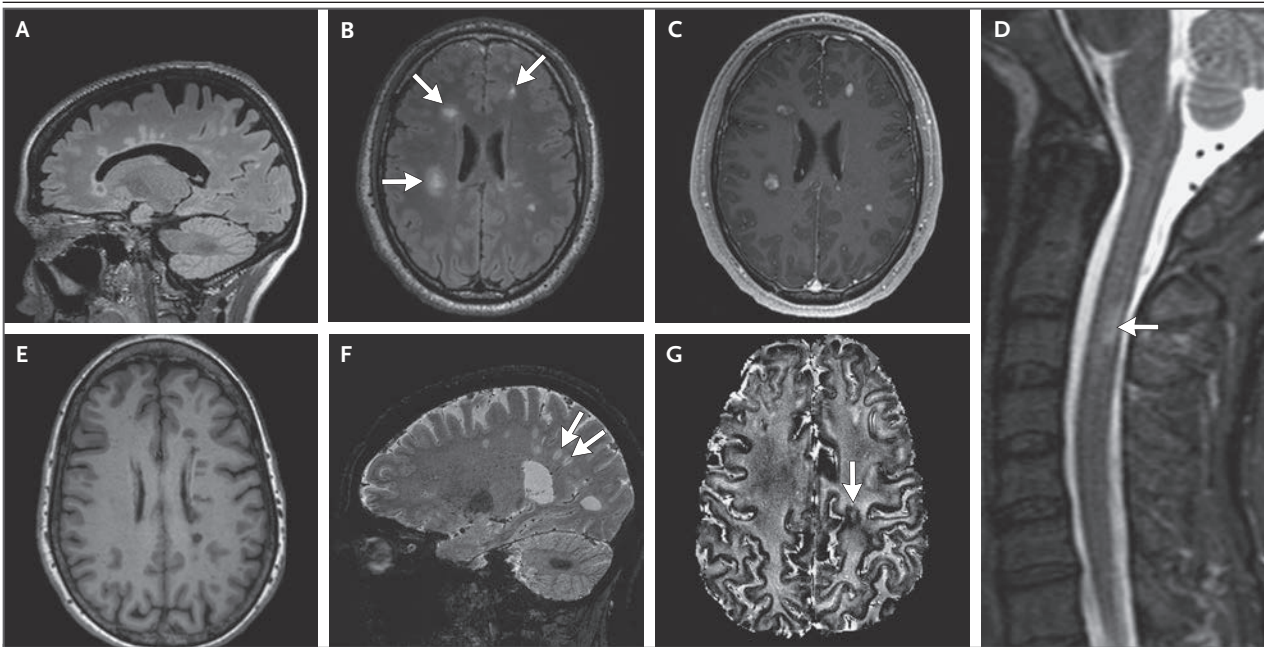


Figure 4. Features of Multiple Sclerosis on MRI.

In Panel A, sagittal T2 FLAIR (fluid-attenuated inversion recovery) shows multiple lesions involving the corpus callosum with their major axis oriented toward the ependymal surface of the ventricle (Dawson's fingers). In Panel B, postcontrast axial T2 FLAIR shows multiple white-matter lesions within the corona radiata bilaterally, consistent with demyelination. Circumferential halos surround the T2 bright cores (arrows); these changes are related to vasogenic edema often associated with blood–brain barrier disruption in acute inflammatory lesions. In Panel C, a T1 postcontrast MPRAGE (magnetization-prepared rapid gradient echo) image from the patient in Panel B shows heterogeneous enhancement within five of the white-matter lesions in the section, indicating active inflammation and demyelination. In Panel D, a sagittal T2 TIRM (turbo inversion recovery magnitude) sequence of the cervical spine shows a short segment with focally increased T2 signal at the dorsal C2–C3 level (arrow). In Panel E, an axial T1 MPRAGE image of a patient with long-standing disease shows lesions that reveal T1 signal intensity that is lower than that of the gray matter; these lesions are defined as “black holes,” radiologic correlates of lesions that are characterized by greater tissue disruption. In Panel F, a sagittal magnitude image from 3D Echo Planar Imaging (3D-EPI) acquisition reveals multiple lesions with a classic perivenular distribution and a positive central vessel sign (arrows). In Panel G, an axial phase image from 3D-EPI acquisition highlights a paramagnetic rim surrounding a nonenhancing lesion within the left centrum semiovale (arrow); paramagnetic rim lesions can represent a radiologic correlate of chronically active plaques. Images courtesy of Simone Sacco.

taper lasting 1 to 2 weeks. Glucocorticoids can speed the tempo of recovery and may have modest longer-term benefits.⁴⁸ Mild attacks are often not treated. Severe attacks that are unresponsive to glucocorticoid therapy can benefit from plasma exchange, typically administered as a course of 5 to 7 exchanges.

PROPHYLAXIS FOR RELAPSING MULTIPLE SCLEROSIS

Many disease-modifying drugs are approved for the treatment of relapsing forms of multiple sclerosis (Fig. 5). Each reduces the frequency of relapses and the development of new lesions seen on MRI. These agents also are indicated for patients with secondary progressive multiple sclerosis who continue to have relapses. Highly effective therapies include monoclonal antibodies directed at CD20-expressing B cells that are in-

fused every 6 months (humanized ocrelizumab, chimeric ublituximab, and rituximab)^{49–51} or subcutaneously every month (fully human ofatumumab)⁵²; the $\alpha 4$ integrin expressed on B and T cells administered by infusion every 4 or 6 weeks (natalizumab),^{53,54} or the pan-lymphocyte-depleting antigen CD52 (alemtuzumab).⁵⁵

Owing to their favorable safety and efficacy profiles, monoclonal antibodies directed against CD20-expressing B cells have emerged as the first-line therapy of choice in many settings.⁵⁶ Anti-CD20 agents inhibit new lesion formation by 99%, and with continued use most patients can expect to remain relapse-free. However, benefits against underlying progression are incomplete, and in some patients there is progression of disability despite treatment. Although each anti-CD20 agent has distinct pharmacokinetic,

pharmacodynamic, and immunogenicity profiles, all provide outstanding control of focal brain inflammation and clinical relapses.⁴⁹⁻⁵² One head-to-head comparison trial found that rituximab was noninferior to ocrelizumab over a 2-year period.⁵⁷ A longer-term observational study showed a higher risk of infections and hospitalizations with rituximab than with ocrelizumab,⁵⁸ and declines in serum immunoglobulin levels also appeared to be steeper with rituximab than with ocrelizumab or ofatumumab.^{58,59} Long-term comparative-effectiveness studies of the different anti-CD20 agents are still needed. Anti-CD20 agents are attractive for use in women who are contemplating pregnancy, because protection against disease activity extends beyond the period of B-cell depletion in peripheral blood and thus treatment given before conception can provide protection throughout gestation.⁶⁰ Extended interval dosing has also been proposed as a more general strategy to reduce the risk and cost of treatment.^{61,62} Another advantage of anti-CD20 drugs is that following cessation of treatment, rebound disease activity is not observed, in contrast with some agents that inhibit lymphocyte trafficking (natalizumab and fingolimod).⁶³

Natalizumab is associated with progressive multifocal leukoencephalopathy, a devastating viral infection of the nervous system, but can be used safely in patients who test seronegative for polyomavirus. The pan-lymphocyte-depleting monoclonal antibody alemtuzumab (anti-CD52) provides long-standing protection after one or two annual infusions; however, the risks of new autoimmunity, cancer, and prothrombotic events have limited the widespread use of this drug. Autologous hematopoietic stem-cell transplantation is an off-label approach that also appears to provide long-term benefit to patients with relapsing multiple sclerosis,⁶⁴ but the risks of immunosuppressant drugs used in conditioning regimens are a safety concern; a late-stage clinical trial is underway (ClinicalTrials.gov number, NCT04047628).

Long-term prospective trials and real-world experience suggest that currently the most effective control of relapsing disease is achieved with initiation of high-efficacy therapy at the earliest possible time.⁶⁴⁻⁶⁶ This practice has largely replaced strategies that began with a less effective therapy and advanced to a more effective one when breakthrough disease occurred. Also underway are randomized clinical trials that should

further clarify the role of high-efficacy therapy as compared with tiered treatment regimens in patients with early disease (NCT03500328 and NCT03535298). Irrespective of the initial agent chosen, in patients who have drug-related side effects or incomplete control of relapse or MRI-detected disease activity, a change in treatment may be warranted.

SECONDARY PROGRESSIVE MULTIPLE SCLEROSIS

As noted, patients with secondary progressive multiple sclerosis who continue to have relapses can be treated with agents approved for relapsing multiple sclerosis. However, many patients with secondary progressive multiple sclerosis do not have relapses, yet they continue to have insidiously progressive disability. Currently, there are no approved treatment options for nonrelapsing secondary progressive multiple sclerosis.

PRIMARY PROGRESSIVE MULTIPLE SCLEROSIS

Ocrelizumab reduced the annual risk of progression by approximately 25% in patients with primary progressive multiple sclerosis.⁶⁷ This effect could translate, on average, to more than 7 years of additional independent ambulation.⁶⁸ Ocrelizumab also preserved arm and hand function in patients with advanced disability who were non-ambulatory.⁶⁹ Thus, although ocrelizumab has been only modestly effective, it should be considered for use in treating most patients with primary progressive multiple sclerosis, especially in younger and less disabled patients who have signs of inflammatory activity on MRI.

SYMPTOM MANAGEMENT

In addition to proper use of disease-modifying therapies, appropriate management of symptoms can improve the patient's quality of life.⁷⁰ Promoting a healthy lifestyle and addressing psychosocial and disease-specific symptoms in the context of a trusting clinician-patient relationship is essential. Low levels of vitamin D should be corrected.⁷¹ Physical-therapy regimens to prevent deconditioning and maintain functional independence are important,⁷² and heat-sensitive patients can use cooling methods such as swimming or exercising in a cool room or wearing cooling garments. Vaccinations that are recommended for use in the general population are safe in patients with multiple sclerosis, with the exception of live vaccines in patients receiving immunosuppressant therapies.⁷³ Advance planning

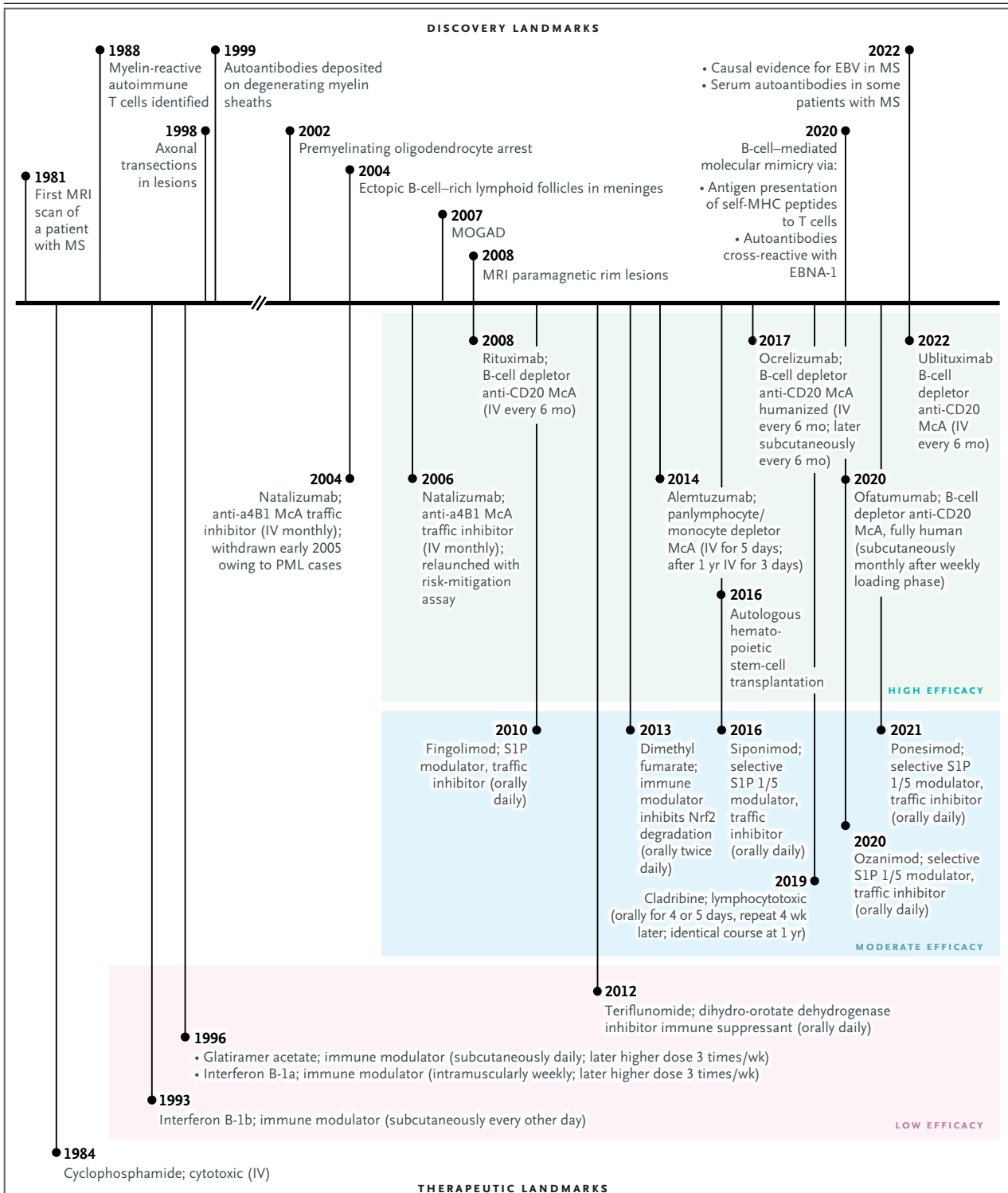


Figure 5. Timeline of Selected Advances in Multiple Sclerosis.

Discovery landmarks are shown above the timeline, and therapeutic landmarks are shown below. Therapies are also grouped according to efficacy tiers, with high efficacy treatments shown in green shading, moderate efficacy treatments in blue shading, and low efficacy in pink shading. IM denotes intramuscular, IV intravenous, McA monoclonal antibody, MHC major histocompatibility complex, MOGAD myelin oligodendrocyte glycoprotein antibody-associated disease, and PML progressive multifocal leukoencephalopathy.

for patients who may become pregnant is recommended before initiating treatment.⁷⁴ A comprehensive treatment plan to address pain, depression, sleep disruption, fatigue, spasticity, multiple sclerosis–related deconditioning, and bladder, bowel, and sexual dysfunction is important. Effective management of coexisting conditions that can accumulate with age and promotion of social engagement could help to improve long-term outcomes. Fatigability and weakness may respond to potassium channel blockers (4-aminopyridine or 3,4-diaminopyridine). Because high-efficacy treatments almost completely eliminate new inflammatory lesions, surveillance MRI scans are often unnecessary in patients whose condition is clinically stable.

CONCLUSIONS AND FUTURE DIRECTIONS

The unmet need is for better control of progressive disability and a halt to neurodegeneration, processes that are thought to be mediated by treatment-resistant B cells, including CD20-negative plasmablasts and plasma cells, cytotoxic CD8 T cells, and neurotoxic glial cells.⁷⁵ An additional complexity is that inflammation in progressive multiple sclerosis is largely sequestered behind an intact blood–brain barrier, a factor that impedes the delivery of many systemically administered drugs, whereas in relapsing multiple sclerosis, B cells in the peripheral circulation are incriminated and the blood–brain barrier is disrupted.

One hypothesis was that higher doses of anti-CD20 monoclonal antibodies might achieve greater benefits against progression, but recent clinical trials with ocrelizumab showed no added benefit.⁷⁶ Targeting CD20-negative plasmablasts with CD19-directed strategies is an alternative approach, either with monoclonal antibodies or chimeric antigen receptor (CAR) T cells that can effectively cross the blood–brain barrier. A case series involving the use of CD19 CAR T cells to treat several autoimmune diseases showed dramatic clinical benefit, with marked reductions of disease-associated autoantibodies and no recurrence of disease activity after the return of B cells to the circulation.⁷⁷ These findings have raised the possibility that the immune system was reset in those cases. The use of CAR T cells directed against other B-cell targets (e.g., B-cell maturation antigen) is also a possible treatment

approach.⁷⁸ There is widespread interest in small-molecule inhibitors of Bruton's tyrosine kinase (BTK) that can block B-cell maturation and immunoglobulin production; BTK is also expressed in microglia (and peripheral macrophages), where it augments proinflammatory signaling. More than a dozen small-molecule BTK inhibitors have been developed, each differing in avidity, specificity for BTK, and central nervous system permeability. Two relatively nonselective BTK inhibitors, evobrutinib⁷⁹ and tolebrutinib,⁸⁰ were not superior to the comparator teriflunomide in relapsing multiple sclerosis. Tolebrutinib was modestly more effective than placebo in nonrelapsing secondary progressive multiple sclerosis⁸¹ but reportedly failed in primary progressive multiple sclerosis. In contrast, the more selective BTK inhibitor fenebrutinib was efficacious in both relapsing and primary progressive multiple sclerosis,^{82,83} and another selective inhibitor, remibrutinib, is currently in late-stage clinical trials (NCT05147220).

Beyond B-cell therapies, efforts to reduce inflammation include targeting the proinflammatory epitope of fibrin, which is deposited in the subendothelia of multiple sclerosis lesions and triggers microglial activation,⁸⁴ as well as approaches that are directed against cytotoxic T cells. Another area of experimental therapeutics is focused on enhancing myelin repair within lesions by promotion of the differentiation of oligodendroglial precursor cells to remyelinate denuded but still-viable axons. A particularly promising target is the M1 muscarinic receptor, and a preliminary clinical trial with the over-the-counter drug clemastine improved visual evoked-potential latencies in patients with relapsing multiple sclerosis, a finding that suggests that remyelination of the visual pathways had occurred.⁸⁵

Considerable interest exists around testing the concept that eradication of Epstein–Barr virus would prevent or ameliorate multiple sclerosis.³¹ A theoretical challenge to any vaccination approach is the possibility that the vaccine might trigger or worsen multiple sclerosis if the epitopes used for immunization cross-react with critical host proteins. Antiviral drugs, or virus-directed T-cell therapy, would presumably be free of this risk.

Finally, it is possible that the initiation of highly effective treatment at the earliest possible time point could disrupt lymphoid niches and block microglial activation responsible for pro-

KEY POINTS

ADVANCES IN MULTIPLE SCLEROSIS

- Multiple sclerosis is a chronic autoimmune disease of the central nervous system that is characterized clinically by acute attacks and gradual progression.
- Myelin is a key target of the autoimmune process, and secondary neurodegeneration is the main contributor to disability.
- Symptoms can include monocular vision loss or double vision, sensory changes, weakness, imbalance, and bladder dysfunction.
- Highly effective disease-modifying treatments now exist for relapsing forms of multiple sclerosis, including monoclonal antibodies that deplete CD20-expressing B cells; these agents have dramatically improved the long-term outlook for patients.
- For pure progressive forms of multiple sclerosis, current disease-modifying therapies have limited efficacy, and more effective treatments to prevent progression and rebuild myelin are needed.
- All patients with multiple sclerosis benefit greatly from effective management of symptoms, including fatigue, spasticity, pain, and bladder dysfunction, as well as avoiding risks of deconditioning, sleep deprivation, and social isolation.

gressive disease. Clinical-trial data indicate that long-term clinical stability, accompanied by normalization of immunoglobulin levels in spinal fluid,⁸⁶⁻⁸⁸ and even resetting of immune repertoires,⁸⁹ is achievable, at least in some patients. With the possibility of presymptomatic diagnosis,³⁵ trials of even earlier therapy could be conceptualized, as undertaken in type 1 diabetes mellitus⁹⁰ and rheumatoid arthritis.⁹¹ With earlier recognition and more effective therapies, future treatment guidelines could evolve away from the need for lifetime therapy and toward early induction treatment followed by maintenance and

off-treatment monitoring regimens to maintain complete suppression of disease activity and, potentially, cure.⁹² Such a vision can only be realized with further advances in the research and science of both the underlying biologic factors of the disease and innovative treatment approaches.

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